

Follow-up: Predicting First-Year Statistics Grades

From Attendance, Honesty, and Quiz Results

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Hypotheses

- **H0:** Attendance, homework completion/timeliness, and academic honesty predict test performance more strongly than attitude toward math.
- **H1:** Seating proximity to higher- or lower-achieving peers affects student performance.
- **H2:** Lower attendance, late homework, digital submission, and low quiz scores are associated with lower final results.

Sample

- 49 students enrolled in the Statistics course at University Goce Delchev (UGD) Shtip.
- Data collected includes attendance, quiz scores, seating positions, and honesty indicators.
- Limitations: Small sample size, potential biases, and measurement constraints.

Experimental Design

- Data collection was unobtrusive to preserve natural behavior.
- Quizzes administered via Google Forms; attendance and seating observed without informing students of specific hypotheses.
- Data aggregated into summary measures to reduce dimensionality.

Feature Engineering

- Variables were summarized (means, indices) for analysis.
- Focused on behavioral, attendance, and seating proximity measures.
- Considered potential confounders such as prior math achievement.

Identifiers & Background

- anonINDX - continuous
ascending student index (1 = first); identifier and a control for random effects.
- GENDER - logical
inferred from name: 0 = male, 1 = female
- HSCH - categorical [1–3]
1 general, 2 economics/law specialized, 3 other
- MJR - categorical [1–6]
1 Finance, 2 Business Economics, 3 Marketing & Management, 4 Accounting & Audit, 5 International Economics & Business, 6 Other
- atitMATH - categorical [1–6]
self-rated attitude to math (1 = awful ... 6 = favorite)

Quiz & Attendance (aggregates)

- $\text{averQIZZ}_{1/2}$ - continuous (0–1)
average quiz score in part 1 / part 2
- $\text{diffQIZZ}_{1/2}$ - continuous (0–1)
change between first quiz and overall quiz average
- $\text{averATND}_{1/2}$ - continuous
average attendance (part 1 / part 2), built from:
 $\text{numP}_{1/2}$
 $\text{numE}_{1/2}$
 $\text{numQ}_{1/2}$

Dishonesty & Homework

- $DISS_1/DISS_2$ - continuous
cumulative dishonesty index ($DISS_2 = DISS_1 + \text{new incidents}$)
components (logical): honLST (falsified signature, +3), honHW (dishonest HW claim, +1), Y1 CHEAT / Y2 CHEAT (suspicious midterm behaviors, +2 each)
- Homework (regular)
numHW_{1/2} - continuous/logical: counts
paperHW_{1/2}, lateHW_{1/2} - continuous/logical: submission format and timeliness
hardHW_{1/2} - categorical (rescaled 0–1): perceived difficulty
- Additional HW additHW_{1/2} - logical; lateADHW_{1/2} - logical (context-specific)

Seating Proximity & Outcomes

- $\text{PROXI25}_{1/2}$ - continuous
fraction of sessions sitting within ± 2 index positions of peers below 25th percentile
- $\text{PROXI75}_{1/2}$ - continuous
fraction of sessions sitting within ± 2 index positions of peers above 75th percentile
proximity computed from per-session seating index (two seats ahead/behind)
- Outcomes $Y1$ / $Y2$ - continuous (0–1, scaled from 0–20 pts):
midterm scores ($Y1$ easier; $Y2$ harder, dynamic assignment)
 $Y1\text{GROUP}$ / $Y2\text{GROUP}$ - group controls

Modeling First Quiz Results

- Regression with demographic and background variables.
- Key variables: No significant predictors; model explains limited variance.
- Notable finding: Prior math achievement likely influential but unmeasured here.

H0: Honesty vs Attitude Toward Math

- **Main finding:** Dishonesty index negatively correlates with exam scores.
- **Attitude toward math:** Weak or no significant effect.
- **Implication:** Academic honesty is a stronger predictor than attitude.

Sample Regression Result

Each point increase in dishonesty index reduces scores by approximately 7.6–11.2 percentage points.

Regression Results for H0

Model Name	1H0	1H0_ar	1H0_ar 2sIs	1H0_reduced	1H0_reduced_ar	1H0_reduced_ar 2sIs	2H0	2H0_ar	2H0_ar 2sIs
Depended Variable :	Y_1	Y_1	Y_1	Y_1	Y_1	Y_1	Y_2	Y_2	Y_2
Sample Size (N)	26	24	24	27	25	25	27	27	24
Endogenous	<i>firstQ1ZZ</i>			<i>firstQ1ZZ</i>			Y_1		
C	0.9994	0.9175	0.9470	0.9123	0.8429	0.7811	0.1203	-0.1095	-0.0942
	0.1908	0.1982	0.2201	0.1885	0.1901	0.2494	0.3583	0.3533	0.7545
	0.0001***	0.0005***	0.0009***	0.0001***	0.0004***	0.0064	0.7407	0.7602	0.9023
firstQ1ZZ or Y_1		0.0215	0.3689		-0.1839	0.3892		0.6230	1.4642
		0.2256	0.5833		0.1847	0.6534		0.3150	0.8655
		0.9255	0.5381		0.3343	0.5597		0.0635*	0.1114
GENDER	0.1162	0.1153	0.1137	0.0973	0.0971	0.0794	0.1440	0.0113	-0.2313
	0.0582	0.0569	0.0619	0.0575	0.0569	0.0744	0.0840	0.1031	0.2455
	0.0631*	0.0636*	0.089*	0.1069	0.1069	0.3020	0.1026	0.9143	0.3611
Y_{1or2} GROUP	-0.1349	-0.1356	-0.1182	-0.1466	-0.1443	-0.1482	0.0563	0.0201	-0.1581
	0.0537	0.0603	0.0707	0.0527	0.0527	0.0668	0.0764	0.0735	0.3141
	0.0232**	0.0423**	0.1183	0.0118**	0.0146**	0.0414**	0.4701	0.7876	0.6221
averATND _{1or2}	-0.1212	-0.0927	-0.1313	-0.0896	-0.0497	-0.0730	-0.0872	-0.0146	0.0115
	0.0469	0.0606	0.0884	0.0453	0.0517	0.0700	0.0748	0.0787	0.1662
	0.02**	0.1504	0.1613	0.0623*	0.3507	0.3127	0.2581	0.8550	0.9457
numHW _{1or2}	-0.0265	-0.0213	-0.0576	0.0025	-0.0057	0.0047	0.2474	0.2234	0.3401
	0.0584	0.0772	0.1006	0.0384	0.0405	0.0525	0.0937	0.0881	0.2317
	0.6563	0.7871	0.5766	0.9483	0.8903	0.9293	0.0161**	0.0207**	0.1628
paperHW _{1or2}	0.1073	0.0773	0.1829						
	0.1155	0.1482	0.2276						
	0.3065	0.6106	0.4362						
lateHW _{1or2}	0.2414	0.2749	0.3808						
	0.1406	0.1522	0.2311						
	0.1053	0.0941*	0.1234						
additHW _{1or2}	0.0134	0.0054	-0.0817	0.0413	0.0833	-0.0432	0.3869	0.2743	0.1499
	0.0922	0.1056	0.1755	0.0919	0.0968	0.1821	0.2049	0.1991	0.5534
	0.8864	0.9601	0.6492	0.6582	0.4023	0.8155	0.0743*	0.1852	0.7902
DISS _{1or2}	-0.0910	-0.0951	-0.0834	-0.0767	-0.0891	-0.0621	-0.1120	-0.0825	-0.0320
	0.0272	0.0274	0.0348	0.0266	0.0268	0.0445	0.0408	0.0409	0.0756
	0.0041***	0.0042***	0.0322**	0.0096***	0.0043***	0.1815	0.013**	0.0587*	0.6785
atitMATH	0.0199	0.0160	0.0306	0.0202	0.0115	0.0271	0.0244	0.0078	-0.0246
	0.0174	0.0238	0.0341	0.0175	0.0204	0.0306	0.0275	0.0269	0.0535
	0.2704	0.5126	0.3860	0.2623	0.5790	0.3894	0.3858	0.7755	0.6529
R^2	0.7098	0.7397	0.6922	0.6585	0.6897	0.5030	0.5740	0.6500	0.3038
Adjusted R^2	0.5466	0.5395	0.4554	0.5327	0.5346	0.2545	0.4170	0.4945	-0.0675
Akaike info criterion	-1.2957	-1.3597	Not calculated	-1.2834	-1.3566	Not calculated	-0.2715	-0.3941	Not calculated
Schwarz criterion	-0.8118	-0.8198	Not calculated	-0.8995	-0.9178	Not calculated	0.1125	0.0379	Not calculated
Log likelihood	26.8439	27.3168	Not calculated	25.3263	25.9577	Not calculated	11.6652	14.3200	Not calculated
F-statistic	4.3481	3.6941	3.1633	5.2337	4.4457	2.7426	3.6568	4.1790	1.9622
Prob(F-statistic)	0.005202***	0.01527**	0.027619**	0.00187***	0.005414***	0.040884**	0.011441**	0.005607***	0.1240
J-statistic	NA	NA	2.8038	NA	NA	0.9201	NA	NA	0.4597
Prob(J-statistic)	NA	NA	0.2461	NA	NA	0.6313	NA	NA	0.7947

H1: Seating Proximity and Peer Effects

- **Key result:** Sitting near higher-achieving peers (PROXI75) positively impacts scores.
- **Effect persists:** Remains significant after controlling for dishonesty.
- **No effect:** Proximity to lower-achieving peers (PROXI25) shows no consistent impact.
- **Mechanism:** Evidence suggests knowledge transfer or social learning, not collusion.

Regression Results for H1 midterm 1

Model Name	1h1	1h1_ar	1h1_ar_2sls	1h1_diss	1h1_diss_ar	1h1_diss_ar_2sls	1h1_diss_int	1h1_diss_int_ar	1h1_diss_int_ar_2sls
Sample Size (N)	46	25	25	46	25	25	46	25	25
Endogenous	<i>firstQ1ZZ</i>			<i>firstQ1ZZ</i>			<i>firstQ1ZZ</i>		
C	0.4553 0.1209 0.0005***	0.5070 0.1839 0.0126**	0.5102 0.1865 0.0131**	0.4809 0.1303 0.0007***	0.7030 0.1667 0.0005***	0.6605 0.1864 0.0023***	0.4182 0.1575 0.0114**	0.7466 0.1821 0.0007***	0.7193 0.2056 0.0028***
firstQ1ZZ		-0.0406 0.2024 0.8431	-0.0873 0.4746 0.8560		-0.1309 0.1717 0.4557	0.1924 0.3476 0.5867		-0.1142 0.1764 0.5260	0.2555 0.3718 0.5013
GENDER	0.0950 0.0621 0.1334	0.1176 0.0870 0.1923	0.1197 0.0893 0.1958	0.0922 0.0628 0.1497	0.1082 0.0727 0.1542	0.0948 0.0805 0.2541	0.0981 0.0637 0.1316	0.1085 0.0739 0.1604	0.0937 0.0839 0.2794
Y1GROUP	-0.1079 0.0408 0.0116**	-0.1224 0.0614 0.0608*	-0.1179 0.0742 0.1284	-0.1103 0.0414 0.011**	-0.1400 0.0517 0.0143**	-0.1684 0.0621 0.0143**	-0.1000 0.0441 0.0288**	-0.1436 0.0528 0.0145**	-0.1771 0.0657 0.0154**
DISS				-0.0115 0.0206 0.5804	-0.0770 0.0254 0.0071***	-0.0687 0.0287 0.0279**	0.0116 0.0382 0.7634	-0.1107 0.0578 0.0725*	-0.1172 0.0651 0.0895*
PROXI ₂₅	0.0039 0.0948 0.9674	0.0125 0.1324 0.9261	0.0087 0.1371 0.9503	0.0050 0.0957 0.9584	-0.0073 0.1108 0.9480	0.0203 0.1238 0.8716	0.0054 0.0962 0.9557	0.0022 0.1136 0.9849	0.0375 0.1309 0.7777
PROXI ₇₅	0.1421 0.0650 0.0345**	0.0597 0.1009 0.5606	0.0650 0.1120 0.5685	0.1404 0.0656 0.0386**	0.0316 0.0848 0.7141	-0.0008 0.0973 0.9933	0.2114 0.1188 0.083*	-0.0418 0.1418 0.7717	-0.1124 0.1701 0.5175
INT(DISS, PROXI ₇₅)							-0.0386 0.0537 0.4763	0.0529 0.0812 0.5234	0.0777 0.0935 0.4177
R ²	0.3578	0.4466	0.4450	0.3628	0.6340	0.5619	0.3711	0.6429	0.5506
Adjusted R ²	0.2952	0.3009	0.2990	0.2831	0.5120	0.4158	0.2744	0.4958	0.3656
Akaike info criterion	-1.0326	-1.0180	Not calculated	-0.9968	-1.3514	Not calculated	-0.9665	-1.2960	Not calculated
Schwarz criterion	-0.8338	-0.7255	Not calculated	-0.7583	-1.0101	Not calculated	-0.6882	-0.9060	Not calculated
Log likelihood	28.7490	18.7248	Not calculated	28.9269	23.8923	Not calculated	29.2299	24.2005	Not calculated
F-statistic	5.7119	3.0664	3.0566	4.5548	5.1960	4.3112	3.8361	4.3720	3.4944
Prob(F-statistic)	0.00095***	0.033978**	0.03437**	0.002216***	0.002924***	0.007239***	0.004221***	0.006108***	0.016547**
J-statistic	NA	NA	4.9859	NA	NA	3.9408	NA	NA	3.0710
Prob(J-statistic)	NA	NA	0.1728	NA	NA	0.2679	NA	NA	0.3808

Regression Results for H1 midterm 1

Model Name	2h1	2h1_ar	2h1_ar_2sls	2h1_diss	2h1_diss_ar	2h1_diss_ar_2sls	2h1_diss_int	2h1_diss_int_ar	2h1_diss_int_ar_2sls
Sample Size (N)	44	44	25	44	44	25	44	44	25
Endogenous	Y_1			Y_1			Y_1		
C	0.1156 0.1221 0.3496	-0.0769 0.1285 0.5528	-0.1483 0.2061 0.4807	0.2847 0.1275 0.0315**	0.0915 0.1217 0.4571	-0.1071 0.2416 0.6629	0.0189 0.1711 0.9128	-0.1784 0.1519 0.2479	-0.0572 0.2317 0.8078
Y_1		0.6756 0.2255 0.0048***	1.3902 0.5230 0.0155**		0.7376 0.1977 0.0006***	1.3499 0.4384 0.0065***		0.7413 0.1833 0.0003***	1.4680 0.5081 0.0102**
GENDER	0.1431 0.0743 0.0615*	0.0616 0.0730 0.4043	-0.1182 0.1248 0.3557	0.1219 0.0689 0.0847*	0.0308 0.0643 0.6349	-0.1122 0.1133 0.3351	0.1377 0.0660 0.0439**	0.0463 0.0599 0.4448	-0.1345 0.1254 0.2983
Y_2 GROUP	0.1039 0.0687 0.1384	0.0712 0.0636 0.2698	-0.0761 0.0767 0.3335	0.0789 0.0639 0.2245	0.0406 0.0562 0.4739	-0.0726 0.0786 0.3677	0.0699 0.0610 0.2594	0.0313 0.0522 0.5520	-0.0655 0.0817 0.4339
DISS				-0.0597 0.0212 0.0076***	-0.0657 0.0184 0.001***	-0.0140 0.0455 0.7621	0.0467 0.0523 0.3770	0.0419 0.0440 0.3466	-0.0613 0.0774 0.4395
PROXI ₂₅	-0.0945 0.0937 0.3196	-0.1079 0.0855 0.2146	0.0654 0.1096 0.5577	-0.0830 0.0865 0.3433	-0.0965 0.0748 0.2050	0.0621 0.1111 0.5833	-0.0283 0.0860 0.7444	-0.0412 0.0724 0.5727	0.0850 0.1203 0.4895
PROXI ₇₅	0.1282 0.0839 0.1347	0.1248 0.0765 0.1110	0.2723 0.0934 0.0089***	0.1603 0.0782 0.0473**	0.1598 0.0675 0.0233**	0.2618 0.1007 0.0181**	0.4878 0.1660 0.0057***	0.4910 0.1396 0.0012***	0.0819 0.2838 0.7765
INT(DISS, PROXI ₇₅)							-0.1322 0.0599 0.0336**	-0.1338 0.0504 0.0117**	0.0901 0.1353 0.5145
R ²	0.2078	0.3591	0.5628	0.3447	0.5238	0.5741	0.4209	0.6018	0.5649
Adjusted R ²	0.1265	0.2748	0.4477	0.2585	0.4466	0.4321	0.3270	0.5244	0.3857
Akaike info criterion	0.0667	-0.0999	Not calculated	-0.0777	-0.3514	Not calculated	-0.1559	-0.4848	Not calculated
Schwarz criterion	0.2694	0.1434	Not calculated	0.1656	-0.0676	Not calculated	0.1280	-0.1604	Not calculated
Log likelihood	3.5327	8.1983	Not calculated	7.7084	14.7316	Not calculated	10.4292	18.6666	Not calculated
F-statistic	2.5568	4.2592	3.8163	3.9980	6.7832	4.3463	4.4829	7.7722	3.7640
Prob(F-statistic)	0.0538	0.0036	0.0146	0.0052	0.0001	0.0070	0.0017	0.0000	0.0121
J-statistic	NA	NA	1.9436	NA	NA	1.8345	NA	NA	3.6342
Prob(J-statistic)	NA	NA	0.5842	NA	NA	0.7662	NA	NA	0.6032

H2: Homework and Performance

- **Result:** No significant linear relationship between attendance, homework orderliness, quiz performance, and perceived difficulty.
- **Possible reasons:** Measurement limitations, small sample size.
- **Future directions:** Better objective measures, larger samples, nonlinear models.

Discussion

- **Main findings:** Peer proximity to higher achievers enhances performance; dishonesty correlates negatively.
- **Limitations:** Small sample, measurement constraints, potential biases.
- **Implications:** Seating arrangements could be a low-cost intervention.
- **Caution:** Results are suggestive; causality not definitively established.

Limitations

- Small sample size and limited external validity.
- Measurement issues: Homework quality, dishonesty proxies.
- Potential observer effects and unmeasured confounders.
- Lack of prior achievement data.

Future Research

- Larger, multi-cohort studies with diverse samples.
- Objective measures: Rubric-scored homework, prior grades.
- Automated participation and attention metrics.
- Use of panel data.
- Explore nonlinear and interaction effects.

Questions and Discussion

I appreciate your attention.

Q & A?